

[Answer the following questions]

- | | P | C | D |
|--|---|---|---|
| 1.a) Given an algorithm with a time complexity of $O(n^2)$ and another with a time complexity of $O(n \log n)$, discuss under what circumstances each algorithm would be preferable in terms of efficiency. Provide real-world examples to support your analysis. | 3 | 2 | 1 |
| b) Discuss the properties of a good algorithm. How do these properties influence the efficiency and effectiveness of an algorithm in solving a given problem? | 3 | 2 | 1 |
| c) Analyze the time complexity of the following algorithm, which computes the factorial of a non-negative integer n : | 4 | 2 | 1 |

```
function factorial(n):
    if n equals 0:
        return 1
    else:
        result = 1
        for i from 1 to n:
            result = result * i
        return result
```

Explain the number of operations performed by the algorithm and justify its time complexity using Big O notation. Consider both the best-case and worst-case scenarios.

OR

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| c) Find the complexity of the followings: | 4 | 2 | 1 |
|---|---|---|---|

```
i) void fun(int n) {
    int i, j, k, count = 0;
    for (i = n/2; i <= n; i++)
        for (j = 1; j + n/2 <= n; j++)
            for (k = 1; k <= n; k = k*2)
                count++;
}
```

```
ii) void fun(int n) {
    if (n <= 1) return;
    int i, j;
    for (i = 1; i <= n; i++)
        for (j = 1; j <= n; j++) {
            printf("Hello\n");
            break;
        }
}
```

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|--|---|---|---|
| 2. a) Given an array of integers: [7, 2, 1, 6, 8, 5, 3, 4], apply the count-sort algorithm to sort this array. Provide a step-by-step sorting process for the algorithm. | 4 | 3 | 1 |
| b) Consider the following recurrence relation:
$T(n) = 2T(n/2) + n$ | 3 | 4 | 2 |

Apply the master theorem and recurrence tree method to analyze the time complexity of this recurrence relation.

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|---|---|---|---|
| c) Starting with the procedure MAX-HEAPIFY, write pseudocode for the procedure MIN-HEAPIFY(A,i), which performs the corresponding manipulation on a min-heap. | 3 | 5 | 2 |
|---|---|---|---|

OR

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| c) Show the steps of the operation Extract-min() on the following min-heap
$A = \{4, 1, 3, 2, 16, 9, 10, 14, 8, 7\}$ | 3 | 5 | 2 |
| 3. a) Find an optimal parenthesization of a matrix-chain product whose sequence of dimensions is
(6, 5, 4, 8, 6). | 6 | 3 | 1 |
| b) Given two strings:
"abcdefgh" & "acbcfghg",
Apply dynamic programming to find the longest common subsequence (LCS) between these two strings. | 4 | 3 | 2 |

OR

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| b) Explain the concept of dynamic programming in the context of solving optimization problems. Illustrate the dynamic programming approach with an example to find the optimal solution. | 4 | 3 | 2 |
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International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE Midterm Examination, Spring 2023
Course Code: CSE 2421 Course Title: Computer Algorithms
Total marks: 30 Time: 90 minutes

[Answer all the questions; In some questions, there might be options;
Figures in the right hand margin indicate full marks.]

1. 4
- a) Find the asymptotic upper bound of the following recurrence by using master method: 4
 $T(n) = 3T(n/3) + f(n)$
where (i) $f(n) = n^2$ and (ii) $f(n) = \lg n$. 3
- b) Find the time complexity of the following function. 3

```
void Duffy (int n)
{
    for (int i = 1; i*i <= n; i++)
        for (int j = 1; j <= i; j = j*2)
            k = k + 1;
}
```

How the time complexity will be changed if $i*i$ is replaced with i ?
- c) Show that the asymptotic lower bound of any polynomial of degree n can be expressed as $\Omega(x^n)$. 3
2. 4
- a) Find the time complexity of the following function? 4

```
Raise (x, n)
if x = 0
    return 1
p = floor(n/2)
if n is odd    return x * Raise(x, p) * Raise(x, p)
if n is even   return Raise(x, p) * Raise(x, p)
```

Can you specify a way to improve the performance of the function?
- b) Show using recursion tree how the worst case time complexity of quicksort algorithm becomes $O(n^2)$. 3
- c) Construct a Heap from the following array and show each step. 3
 $A = (5, 9, 3, 14, 7, 12, 11, 8, 4, 10, 6)$.
- OR
- Show how you can sort the numbers from the following Heap A. Illustrate each step.
 $A = (10, 9, 8, 7, 6, 5, 4, 3, 2, 1)$.
3. 4
- a) Find an optimal parenthesization of a matrix-chain product whose sequence of dimensions is $(5, 5, 6, 6, 8)$. 4
- OR
- Find a Longest Common Sequence of the following two sequences using dynamic programming and show the steps.
 $X = (0, 1, 1, 0, 1, 0, 1)$
 $Y = (1, 0, 0, 0, 1, 1)$
- b) What is optimal substructure? Show that the matrix-chain-multiplication problem has this property. 3
- OR
- What is overlapping subproblems? Show that the longest common subsequence problem has this property.
- c) Show the operation of the Counting-sort on the following array. 3
 $A = (6, 4, 8, 4, 6, 2, 6, 4, 8, 2, 7)$

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Time: 90 minutes

[Answer all the questions; in some questions, there might be options;
Figures in the right hand margin indicate full marks.]

1. CO . DL
- a) Let $K = ((X \text{ MOD } 2) + 1) * 2$, where X is the last digit of your ID. Now, find the asymptotic upper bound of the following recurrence by using master method. CO4 N 4
- $T(n) = KT(n/2) + n$
- Verify your answer through recursion tree method.
- b) Compare the time complexity of the following two functions. Which one runs faster. CO4 E 3
- ```
void Chip (int n)
{
 for (int i = 1; i <= n; i++)
 for (int j = 1; j <= i; j = j*2)
 k = k + 1;
}
```
- ```
void Dale (int n)
{
    for (int i = 1; i <= n; i = i*2)
        for (int j = 1; j <= i; j++)
            k = k + 1;
}
```
- c) Show that the asymptotic upper bound of the following function is not $O(n^2)$: $An^3 + Bn^2 + C$. Assume that A , B , and C are any constants. CO4 N 3
- OR
- Show that the asymptotic lower bound of the following function is not $\Omega(n^2)$: $An^3 + Bn^4 + Cn^5$. Assume that A , B , and C are any constants.
- 2.
- a) Prove that we can convert an array into a heap in $O(n)$ time. CO4 N 4
- b) Write down the recurrence equation for the running time of quicksort algorithm in which partition always produces a 9-to-1 proportional split. Solve the recurrence using recursion tree and show that in this case running time is $O(n \lg n)$. CO3 N 3

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Total marks: 30

Time: 90 minutes

[Answer all the questions; in some questions, there might be options;
Figures in the right hand margin indicate full marks.]

1. Find the time complexity of the following function? 4
- 1.a)

```
Fib (n)
    if (n = 0 or n = 1)    return 1
    return Fib(n-1) + Fib(n-2)
```

 How will it be changed if the function is written as follows?


```
Fib_tail (n, a=0, b=1)
    if (n == 0)    return a
    if (n == 1)    return b
    return Fib_tail (n-1, b, a+b)
```

 Here, a=0, b=1 are default arguments.
- 1.b) Find the time complexity of the following function. 3
- ```
void Tweety (int n)
{
 for (int i = 1; i <= n; i++)
 for (int j = 1; j <= i*i; j++)
 k = k + 1;
}
```

 How the time complexity will be changed if  $i*i$  is replaced with  $i$ ?
- 1.c) Find the asymptotic bound of the following recurrence by using master method: 3
- $T(n) = 4T(n/k) + n^2$   
where (i)  $k = 4$  and (ii)  $k = 2$ .
- 2.a) Show that the lower bound of any comparison based sort is  $\Omega(n \lg n)$ . 4
- OR
- Show that a heap can be constructed from an array in  $O(n)$  time. 3
- 2.b) Show the operation of the Radix-sort on the following array. 3
- $A = \{673, 424, 527, 639, 247, 826, 974, 428, 356, 652, 713\}$
- 2.c) Show the steps of sorting the following list of numbers using Quicksort. Show steps of each partition operation. 3
- $A = \{15, 9, 5, 2, 3, 6\}$
- OR
- Show how you can sort the numbers of the following min-heap A. Illustrate each step. 4
- $A = \{1, 2, 3, 6, 8, 11, 13, 15, 17, 21, 23\}$ .
- 3.a) Find an optimal parenthesization of a matrix-chain product whose sequence of dimensions is  $\{3, 7, 9, 5, 8\}$ . 3
- 3.b) What is optimal substructure? Show that the longest common subsequence problem has this property. 3
- OR
- What is overlapping subproblems? Show that the matrix-chain-multiplication problem has this property.
- 3.c) How many ways you can parenthesize a chain of 7 matrices? Show the steps of calculation. 3
- Also, find how many distinct subproblems you will encounter if you want to parenthesize a chain of 7 matrices optimally for multiplication.

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B. Sc. in CSE Mid-term Examination, Autumn 2023

Course Code: CSE 2423

Course Title: Database Management



Edit

Annotate

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Convert

All

Bismillahir Rahmanir Rahim  
International Islamic University Chittagong  
Department of Computer Science & Engineering  
B.Sc. in CSE, Mid-term Assignment, Autumn 2021  
**Course Code: CSE-2421 Course Title: Computer Algorithms**  
Total Marks: 21 Time: 2.5 hours + 30 minutes

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Answer all three; all questions have the same marks.

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For all questions, assume **L2** denotes 1+ second last digit of your ID, **L1** denotes 1+ last digit of your ID and **L#** denotes the 1+ last two digits of your ID;  
Example: For C171050, **L2**=6, **L1**=1 and **L#**=51.

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- 1.a) Define: Algorithm. Briefly describe the Properties of good algorithm. 2
- b) Express the following recurrence in  $\Theta$  notation:  $T(n) = (L1) T(\frac{n}{L2}) + n^5$  2
- c) **Transpose(A, L#)** 3  
    for **i=1** to **L# - 1**  
        for **j=i+1** to **L#**  
            **t = A[i,j]**  
            **a[i,j] = A[j,i]**  
            **A[j,i] = t**  
Consider the above pseudocode. Mention how many times each line executes; express in asymptotic notation. Express the running time of **Transpose** in asymptotic notation.
- 2.a) Use Counting sort algorithm to sort the following list: **5, L2, L2, 26, L1, 34, L1, 7, 22.** 2
- b) Perform Heap sort operation on the following list of data: **41, L1, 35, L#, 45, L2, L1, L2.** 2
- c) Use Quick sort algorithm to sort the following list: **L#, L2, L1, 26, L2, L1, 3, L1, 2.** 3
- 3.a) What are the differences between Dynamic Programming and Divide and Conquer approach? 1
- b) Determine the LCS of two strings  $S_1 = \{5, L1, 0, L1, 0, 5, L2, 3\}$  and  $S_2 = \{L1, 5, L2, 5, L1, 0, 3, 5, 0\}$ . 3
- c) Find the values of m-table and s-table of a matrix-chain product whose sequence of dimensions is  $\{L1, L2, 5, L#, L2\}$ . 3